

Exponent Laws with Variable Bases

MA.8.AR.1.1

I can multiply and divide monomials, and raise a monomial to a power, by applying the laws of exponents to variable bases.

Yesterday the bases were all numbers.

Today the base is a letter. Every law still works.

Warm-Up

On your own. Four minutes. No notes.

1. Find the value: $(2^3)^2 \cdot 2^{-4}$
2. Write as one power: $\frac{6^7}{6^5}$
3. $\sqrt{50}$ is between which two whole numbers?
4. Write with an exponent: $y \cdot y \cdot y \cdot y \cdot y$

Warm-Up

Answers.

1. Find the value: $(2^3)^2 \cdot 2^{-4}$ **4**
2. Write as one power: $\frac{6^7}{6^5}$ **$6^2 = 36$**
3. $\sqrt{50}$ is between which two whole numbers? **7 and 8**
4. Write with an exponent: $y \cdot y \cdot y \cdot y \cdot y$ **y^5**

Notes

Copy both lines.

I. Product of powers — the base is a letter now

$$x^2 \cdot x^3 = (x \cdot x)(x \cdot x \cdot x) = x^5 \qquad x^m \cdot x^n = x^{m+n}$$

$$3x^2 \cdot 5x^4 = (3 \cdot 5)(x^2 \cdot x^4) = 15x^6$$

Multiply the coefficients. Add the exponents of like bases.

Different variables stay separate: $x^2y \cdot x^3y^4 = x^5y^5$

Notes

II. Power of a power, power of a product

$$(x^3)^4 = x^3 \cdot x^3 \cdot x^3 \cdot x^3 = x^{12} \qquad (x^m)^n = x^{mn}$$

$$(2y^2)^3 = 2^3 \cdot (y^2)^3 = 8y^6 \qquad (15xy^2)^3 = 3,375x^3y^6$$

Power of a power: multiply the exponents. Power of a product: raise every factor.

Notes

III. Quotient of powers — dividing monomials

$$\frac{x^7}{x^2} = \frac{x \cdot x \cdot x \cdot x \cdot x \cdot x \cdot x}{x \cdot x} = x^5$$

$$\frac{x^m}{x^n} = x^{m-n}$$

$$\frac{12a^5}{4a^2} = 3a^3$$

$$\frac{m^6n^4}{m^2n^4} = m^4n^0 = m^4$$

Divide the coefficients. Subtract the exponents of like bases.

Example 1

Multiplying two monomials.

$$(4x^3y)(-2x^2y^5)$$

Write an equivalent expression as a single monomial.

Example 1

Worked.

$$(4x^3y)(-2x^2y^5) = (4)(-2) \cdot x^3x^2 \cdot y^1y^5 \quad \text{regroup: coefficients, } x\text{'s, } y\text{'s}$$
$$= -8 \cdot x^5 \cdot y^6 = -8x^5y^6 \quad \text{multiply coefficients; add exponents}$$

Answer: $-8x^5y^6$

Your Turn

Same steps, your numbers. Boards up when done.

$$(5a^2b^4)(3ab^2)$$

Your Turn

Answer.

$$(5a^2b^4)(3ab^2)$$

$$5 \cdot 3, a^2 \cdot a^1, b^4 \cdot b^2$$

$$\textbf{Answer: } 15a^3b^6$$

Example 2

A power of a product, then a quotient.

$$\frac{(3m^2n)^3}{9m^4n}$$

Write an equivalent expression as a single monomial.

Example 2

Worked.

$$(3m^2n)^3 = 27m^6n^3 \quad \text{power of a product: raise all three factors}$$

$$\frac{27m^6n^3}{9m^4n} = 3 \cdot m^{6-4} \cdot n^{3-1} = 3m^2n^2 \quad \text{divide coefficients; subtract exponents}$$

Answer: $3m^2n^2$

Your Turn

Same steps, your numbers. Boards up when done.

$$\frac{(2p^3)^4}{8p^5}$$

Your Turn

Answer.

$$\frac{(2p^3)^4}{8p^5}$$

$$16p^{12} \text{ over } 8p^5$$

Answer: $2p^7$

Whiteboards • Question 1 of 9

Boards up on three.

$$x^5 \cdot x^8$$

Answer it.

Write as a single power.

Whiteboards • Question 1 of 9

Answer.

$$x^5 \cdot x^8$$

like bases: add the exponents

Answer: x^{13}

Whiteboards • Question 2 of 9

Boards up on three.

$$(y^4)^3$$

Answer it.

Write as a single power.

Whiteboards • Question 2 of 9

Answer.

$$(y^4)^3$$

power of a power: multiply

Answer: y^{12}

Whiteboards • Question 3 of 9

Boards up on three.

A cube has edge $2x^3$ centimeters.

Its surface area is $24x^6$ square centimeters.

Write the cube's volume as a single monomial.

Answer it.

Write as a single monomial.

Whiteboards • Question 3 of 9

Answer.

A cube has edge $2x^3$ centimeters.

Its surface area is $24x^6$ square centimeters.

Write the cube's volume as a single monomial.

$(2x^3)^3$: raise the 2 as well

Answer: $8x^9$

Whiteboards • Question 4 of 9

Boards up on three.

$$(3a^2b)(4ab^5)$$

Answer it.

Write as a single monomial.

Whiteboards • Question 4 of 9

Answer.

$$(3a^2b)(4ab^5)$$

$$3 \cdot 4, a^2 \cdot a, b \cdot b^5$$

$$\textbf{Answer: } 12a^3b^6$$

Whiteboards • Question 5 of 9

Boards up on three.

$$(5m^3)^2$$

Which expression is equivalent?

A. $25m^6$

B. $10m^6$

C. $25m^5$

D. $5m^6$

Answer it.

Whiteboards • Question 5 of 9

Answer.

$$(5m^3)^2$$

Which expression is equivalent?

A. $25m^6$

B. $10m^6$

C. $25m^5$

D. $5m^6$

Answer: A — $25m^6$

Whiteboards • Question 6 of 9

Boards up on three.

$$\frac{12x^8}{3x^2}$$

Which expression is equivalent?

A. $4x^6$

B. $4x^4$

C. $9x^6$

D. $4x^{10}$

Answer it.

Whiteboards · Question 6 of 9

Answer.

$$\frac{12x^8}{3x^2}$$

Which expression is equivalent?

A. $4x^6$

B. $4x^4$

C. $9x^6$

D. $4x^{10}$

Answer: A — $4x^6$

Whiteboards • Question 7 of 9

Boards up on three.

$$\frac{m^6 n^4}{m^2 n^4}$$

Answer it.

Write as a single monomial.

Whiteboards • Question 7 of 9

Answer.

$$\frac{m^6 n^4}{m^2 n^4}$$

$$m^4 \cdot n^0, \text{ and } n^0 = 1$$

Answer: m^4

Whiteboards • Question 8 of 9

Boards up on three.

Find the value of n :

$$x^3 \cdot x^n = x^{11}$$

Answer it.

Which law applies?

Whiteboards • Question 8 of 9

Answer.

Find the value of n :

$$x^3 \cdot x^n = x^{11}$$

$$3 + n = 11$$

Answer: $n = 8$

Whiteboards • Question 9 of 9

Take your time. Boards up when you have written it.

Rachel writes $(15xy^2)^3$ as $15 \cdot 15 \cdot 15 \cdot x \cdot x \cdot x \cdot y \cdot y \cdot y \cdot y \cdot y \cdot y$.

Justina writes it as $3,375x^3y^6$. Who is correct? Explain.

Write your answer in sentences.

Say what each of them did.

Whiteboards • Question 9 of 9

Answer.

Rachel writes $(15xy^2)^3$ as $15 \cdot 15 \cdot 15 \cdot x \cdot x \cdot x \cdot y \cdot y \cdot y \cdot y \cdot y \cdot y$.

Justina writes it as $3,375x^3y^6$. Who is correct? Explain.

both — Justina's is Rachel's product, multiplied out

Answer: Both are correct; $15^3 = 3,375$, $x \cdot x \cdot x = x^3$, and six y 's make y^6 .

Independent Set

Six questions. On your own, in silence.

1. Six questions. Work down the page.
2. Show the step that does the work, not just the answer.
3. Silence until the six minutes are up.

IXL

Last five minutes.

1. Open IXL and start today's skills.
2. Work on paper where the question needs work. The answer box does not show it.
3. Every skill listed is required, to a SmartScore of 67. Due at the start of the next class.

Today's skills — all required

Multiply powers: variable bases — TR9

Powers of a power: variable bases — 2CU

Divide monomials with positive exponents — C8L